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Utilities

```
+getEuclideanDistance(x: int, y: int, x2: int, y2: int): double { static }
+clamp(val: double, min: double, max: double): double { static }
+getRadiusDistance(a: double, b: double): double { static }
```

Entity

```
-image: Image
-x: double
-y: double
-radius: double
-visible: boolean = true
-speed: int
-health: int

+moveDown(): void
+moveLeft(): void
+moveRight(): void
+reset(): void
+setVisible(visible: boolean): void
+render(): void
```

Collied

```
-SMOKE: Image (final)
-smokeX: int
-smokeY: int
-timeOut: int
-moveAwayFrame: int
-collidedEntity1: Entity
-collidedEntity2: Entity
-SMOKE_DURATION: int (final)
-fireX: int
-fireY: int
-FIRE_DURATION: int (final)
-FIRE: Image (final)
-SPEED_FIRE: int (final)

+hasCollied(): boolean
+renderCollision(): void
+renderAftermath(): void
+renderFire(): void
```

Coin

```
-DURATION: int (final)
-powerIsActive: boolean = false
-coinPower = 0;

+updateCoinPower(duration: int): void
+renderCoinPowerFrame(for Font, x: double, y: double)
+collidedWithCoin(coin: Coin): void
```

Driver

```
-BLOOD: Image (final)
-isInvincible: boolean = false
-RUN_SPEED: int (final)

+updateHealth(value: damage: int): void
+renderHealth(value: font: Font, x: int, y: int): void
+isInvincible(): void
+run(): void

+takeDamage(entity: Damageable): void
+hasCollied(): boolean

+handleEjection(entity: Ejectable): void
```

<<interface>> Ejectable

<<interface>> Ejection

```
+handleEjection(entity: Ejectable): void
```

HomeScreen

```
-MESSAGE: String (final)
-MESSAGE_FONT: Font (final)
-GAME_LOST: String (final)
-ENTER: String (final)
-ENTER_FONT: Font (final)
-HOME: Image (final)

+render(input: Input): void
```

GameEndScreen

```
-BG: Image (final)
-GAME_WON: String (final)
-GAME_LOST: String (final)
-HIGHEST_SCORE_FONT: Font (final)
-STATUS_FONT: Font (final)
-SCORE_FONT: Font (final)
-top_score: Score[]
-score: Score[]
-isWon: boolean

+ getTopScore(): Score[]
+ populateScore(filename: String)
+ update(input: Input): void
+ isWon(boolean isWon): boolean
+ setWon(boolean isWon): void
```

Road

```
- BG_SUNNY: Image (final)
-BG_RAIN: Image (final)
-speed: int { final }
-roadLaneCenter: int[]
-maxY: int (final)
-currentWeather: String

+moveBG(): void
+setWeather(weather: String): void
```

<<interface>> Invincible

InvinciblePower

```
-DURATION: int (final)

+checkCollisionWithPower(entity: Entity): boolean
+effecting(taxi: Car): void
```

Passenger

```
-priority: int
-pickedUp: boolean
-droppedOff: boolean
-expectedValue: double
-isExpectedValueCalculate: boolean = false
-hasUmbrella: boolean
-health: double
-collisionTimeOut: int
-BLOOD: Image (final)

+renderPriority(font: Font, rate: double, ratePerY: double)
+moveToFlag(speedX: int, speedY: int)
+renderPassenger(font: Font, rate: double, ratePerY: double)
+expectedValue(rate: double, ratePerY: double)
+calculateExpectedValue(rate: double, ratePerY: double)
+render(font: Font, ratePerY: double, rate: double)
+takeDamage(entity: Damageable): void
+hasCollied(): void
+handleEjection(entity: Ejectable): void
+renderHealth(): void
```

GameScreen

```
-coinList: List<Coin>
-invincible: List<Invincible>
-passengerList: List<Passenger>
-weather: String[]
-tripEndFlagList: List<tripEndFlag>
-otherCar: String[]
-lane: int[]
-FONT_GAME: Font (final)
-FONT_PASSSENGER: Font (final)
-PASSENGER_SPEED_X: int (final)
-PASSENGER_SPEED_Y: int (final)
-RATE: int (final)
-POWER_X: int (final)
-POWER_Y: int (final)
-TRIP_STATUS_X: int (final)
-TRIP_STATUS_Y: int (final)
-LAST_TRIP_X: int (final)
-LAST_TRIP_Y: int (final)
-RATE_PER_Y: int (final)
-duration: double
-COIN_DURATION: double (final)
-penaltyRate: double
-gamePlayStrings: String[]
-passengerStrings: String[]
-gamePlayValue: String[]

+moveObject(input: Input)
+drivingCar(input: Input)
+renderGameScreen(input: Input)
+gameLoop(): void
```

ShadowTaxi

```
-GAME_PROPS: Properties (final)
-MESSAGE_PROPS: Properties (final)

+update(input: Input): void
+main(args: String[]): void { static }
```

PlayerInfoScreen

```
-BG_PLAYER_INFO: Image (final)
-INFO_FONT: Font (final)
-NAME_ENTER: String (final)
-NAME_START: String (final)
-PLAYER_NAME_Y: int (final)
-PLAYER_START_Y: int (final)
-INFO_WIDTH: double (final)
-START_WIDTH: double (final)
-setName: String = ""

+nameMaker(input: Input)
+renderPlayerInformation(input: Input)
```

Car

```
-DAMAGE_TAXI: Image (final)
-collisionDamage: int
-isDamage: boolean = false
-oldX: int
-oldY: int
-slop: boolean
-SPEED: int (final)
-healthValueX: int
-healthValueY: int
-HEALTH_VALUE_FONT: Font (final)
-CAR_DAMAGE: int (final)

+renderHealth(font: Font): void
+takeDamage(entity: Damageable): void
+hasCollied(): boolean
+infiicDamage(attack: Damageable): void
+checkInvincibility(): void
+newTaxi(): Car
+isInvincible(duration: int): void
+permantlyDamage(): void
+collialbe(car: Car): boolean
```

TripEndFlag

<<interface>> Damageable

<<interface>> Attack

```
+infiicDamage(attack: Damageable): void
```

Trip

```
-earning: double
-tripMaxFrame: int
-penalty: double
-priority: int
-isDripped: boolean
-isFinish: boolean

+getEarnedPay(): double
+pickedUp(passenger: Passenger, taxi: Car)
+dropOffPassenger(FONT_GAME: Font, passengerString: String[], tripStatusX: int, tripStatusY: int)
+hasPassenger(): boolean
+updatePay(): void
+renderPay(): void
+renderTrip(font: Font, tripStatus: String[], tripStatusX: int)
+renderLastTrip(font: Font, tripStatus: String[], tripStatusX: int)
+checkPenalty(passenger: Passenger, penalty: double)
+updateFrame(font: Font, gameText: String[], gameValue: double[], duration: int)
+setTripMaxFrame(): void
+caculateFinalEarning(earning: double, double: penalty): double
+priorityCalculation(ratePerY: int, priority: int, rate: int, endY: int, startY: int): double
+renderMinimumHealth(): void
```

<<interface>> TakeDamage

```
+takeDamage(entity: Damageable): void
+hasCollied(): boolean
```

EnemyCar

```
-EMEMY_CAR: Image (final)
-DAMAGE_POINT: int (final)
-x: int
-y: int
-FIRE_BALL: Image (final)
-FIRE_BALL_SPEED: int (final)
-FIRE_BALL_DAMAGE: int (final)

+infiicDamage(attack: Damageable): void
```

OtherCar

```
-CAR_1: Image (final)
-CAR_2: Image (final)
-newX: int
-newY: int
-SPEED: int (final)
-isDamage: boolean = false

+createCar(): void
+newCar(): otherCar
+permantlyDamage(): void
+infiicDamage(attack: Damageable): void
```